

The Magic Express: Speeding Up Lists & Sorting



The Hook - Sieve vs Hand-picking

**Imagine picking Alphonso mangoes one by one... or using a Magic Sieve that picks only ripe ones!
List comprehension = Magic Express – fast and smart!**



List Comprehension



**A shortcut to make new lists quickly
– one line instead of many!**

Syntax: [x for x in list]

Example:

**festivals = ["Diwali", "Holi", "Eid",
"Pongal"]**

Old Way vs New Way

Old way

Old way

```
f/long —{  
  python for-loop  
code:  
sostun 'fk(l}  
}
```

Magic Express

```
list list  
comprehension  
code:  
et:tc
```

**Compare the traditional
loop method with the Magic
Express list comprehension
- same result, fewer lines!**

Hands-on Filtering

**Let's pick only the
odd-numbered
houses – quick and
easy!**



```
▼.index(nth) {  
  number(2d, eal(yy1y:-1))0lyb(im nwanall))}  
  loplin:110og|}  
  
  f.texts(#6d, clldc.ist } {  
    na.fndue=caid-dcurrs canlmauehonsort():  
    na.iordl, value(a)) }  
  eal.rysin:houses() >  
  ● < . Hands-on
```



Sorting - What Is It?

**Sorting is arranging things nicely –
like a librarian arranging books or a
Kirana store sorting spices by price!**

**Ascending = climbing temple stairs
up**

Descending = coming down

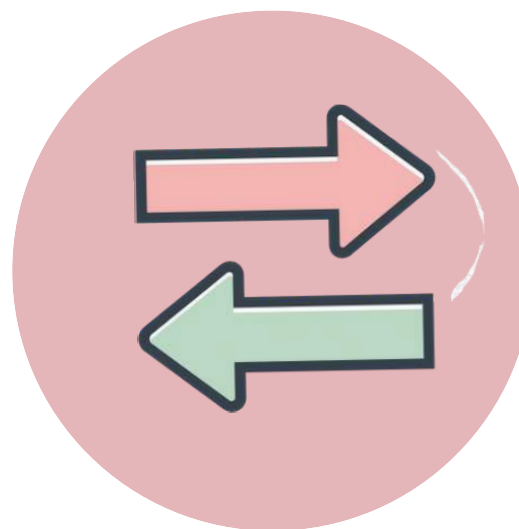


`.sort()` vs `sorted()`

ABC

S O R T T E X T

`.sort()` changes your list directly!
`sorted()` gives you a new sorted list!
`festivals = ["Holi", "Diwali", "Eid"]`
`festivals.sort()` # Now: `["Diwali", "Eid", "Holi"]`
`new_list = sorted(["Pongal", "Bihu", "Onam"])`
`new_list: ["Bihu", "Onam", "Pongal"]`



Sorting Numbers - Cricket Scores Example



Gali match scores: [12, 7, 19, 5, 22]

Ascending: `sorted(scores)` → [5, 7, 12, 19, 22]

Descending: `sorted(scores, reverse=True)` → [22, 19, 12, 7, 5]

Who scored the highest? Let's find out easily!

Sorting Coins

coins = [10, 1, 5, 2, 1]

Ascending: sorted(coins) → [1, 1, 2, 5, 10]

**Descending: sorted(coins,
reverse=True) → [10, 5, 2, 1, 1]**

**Sorting helps organise things super
fast!**



Why Use List Comprehensions & Sorting?

- ✨ Cleaner code = easier to read
- ⚡ Faster to write = more time for fun!
- 🎯 Less mistakes = better magic powers



One-Line Wonder Challenge

Can you write this in one line using list comprehension?

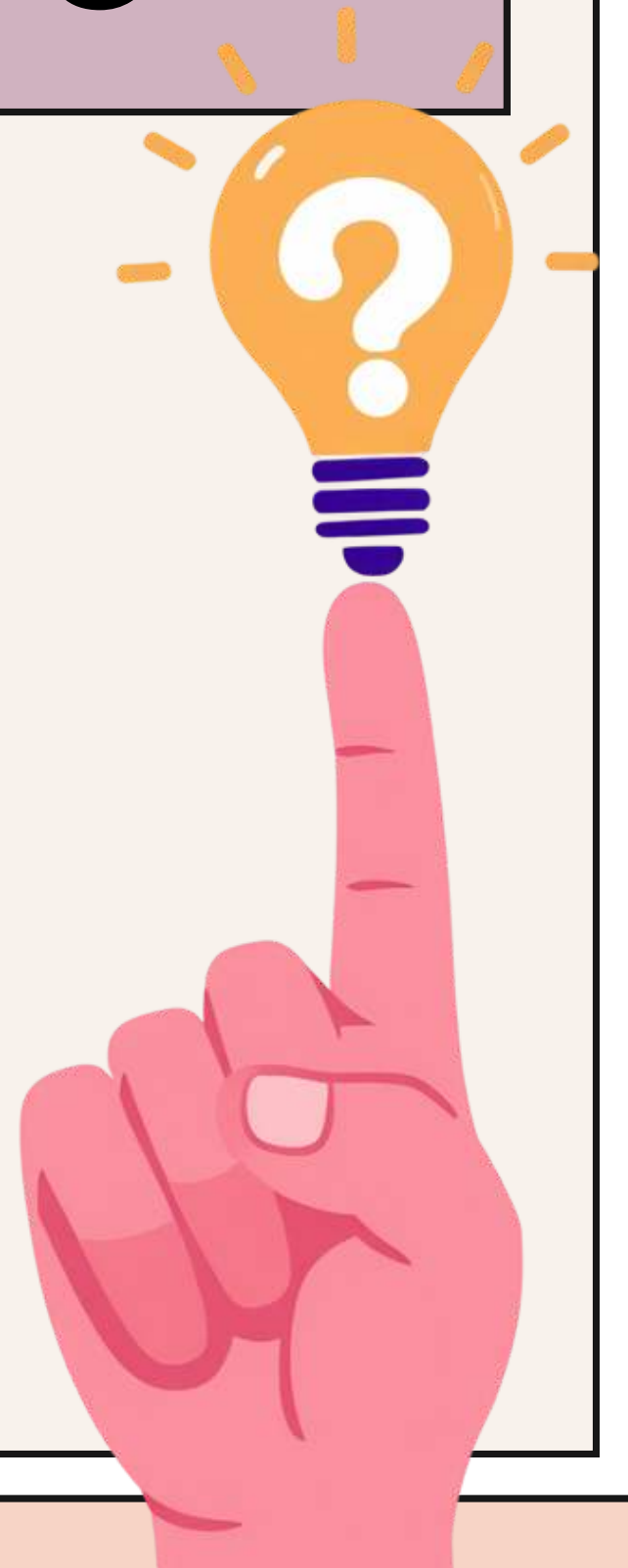
```
numbers = [1, 2, 3, 4, 5]
```

```
doubled = []
```

```
for n in numbers:
```

```
    doubled.append(n * 2)
```

Think and try your Magic Express shortcut!



Summary

List Comprehension = Magic Express shortcut

Filter lists with conditions in one line

 **Sort ascending (climbing up) & descending (coming down)**

🌟 `.sort()` changes your list, `sorted()` gives a new one

You've mastered the Magic Express ride!



Thank You & Keep Coding!

**You are now coding ninjas! Keep practicing your Magic Express skills.
See you in the next session!**

